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Original Article

Modelling high-resolution spatio-temporal wind with deep echo state networks and stochastic partial differential equations

Kesen Wang[1], Minwoo Kim[2], Stefano Castruccio[3] and Marc G. Genton[1]

[1]Statistics Program, King Abdullah University of Science and Technology, Thuwal 23955-6900, Saudi Arabia
[2]Department of Statistics, Pusan National University, Busan 46241, South Korea
[3]Department of Applied and Computational Mathematics and Statistics, University of Notre Dame, Notre Dame, IN 46556, USA

Address for correspondence: Marc G. Genton, Statistics Program, King Abdullah University of Science and Technology, Thuwal 23955-6900, Saudi Arabia. Email: marc.genton@kaust.edu.sa

Abstract

In the past decades, clean and renewable energy has gained increasing attention due to a global effort on carbon footprint reduction. In particular, Saudi Arabia is gradually shifting its energy portfolio from an exclusive use of oil to a reliance on renewable energy, and, in particular, wind. Modelling wind for assessing potential energy output in a country as large, geographically diverse and understudied as Saudi Arabia is a challenge which implies highly non-linear dynamic structures in both space and time. To address this, we propose a spatio-temporal model whose spatial information is first reduced via an energy distance-based approach and then its dynamical behaviour is informed by a sparse and stochastic recurrent neural network (Echo State Network). Finally, the full spatial data is reconstructed by means of a non-stationary stochastic partial differential equation-based approach. Our model can capture the fine scale wind structure from a high-resolution simulation and produce more accurate forecasts of both wind speed and energy in lead times of interest for energy grid management.

Keywords: echo state network, spatio-temporal model, stochastic partial differential equation, support points, wind energy

1 Introduction

The constant increase in the carbon footprint in the past decade has caught the attention of the world. Emerging countries such as Saudi Arabia, whose economy has been so far almost exclusively focussed on non-renewable energies, announced 'Vision 2030', a comprehensive blueprint outlining measures to transition into an economy more focussed on renewable energy. Saudi Arabia's economic development in the past century has been almost exclusively driven by its large oil reserves, which comprise a quarter of the world's proven reserves and currently produce more than a tenth of the annual global oil output (Nakov & Nuno, 2013). To diversify the country's economy, recent efforts have focussed on reducing its excessive reliance on fossil fuels and investing in renewable energy. Indeed, 'Vision 2030' has explicitly laid out a plan to construct wind turbines and contribute 16 GW to the domestic energy demand. Since there is currently little to no infrastructure for wind power in Saudi Arabia, understanding the local spatio-temporal structure of wind to locate and then forecast the most suitable construction sites is a strategic priority. In wind energy markets for developed countries such as the United States, numerical weather models such as the High Resolution Rapid Refresh model (HRRR, Dowell et al., 2022) provide a

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continuous update of the wind (among other variables). In emerging countries such as Saudi Arabia, there is no cyberinfrastructure for a systematic and continuous update of the wind speed at high spatio-temporal resolution. Additionally, it is not possible to rely on observational data, which are too sparse and coarse to provide a country-level mapping of wind (Crippa et al., 2021; Giani et al., 2020). To address this, Giani et al. (2020) provided the first *ad hoc* high resolution simulation performed over the country as no other existing dataset could provide the same level of information, and used it to design a comprehensive plan to locate wind turbine sites that account for energy potential as well as associated maintenance costs. Additional work has focussed on these sites to provide a sensitivity study of potential energy production against varying climate conditions (Zhang et al., 2021) or disruption events associated with extreme conditions causing turbine shut-offs or failures (Chen et al., 2021). The said studies provide a first assessment to support the country's plan for a sustainable wind energy portfolio; however, contingent political, economic, and societal circumstances may be such that the optimal sites may not be eventually chosen for wind farming. As such, given the current lack of a definitive choice of wind farm sites, a reliable forecast approach is still necessary for wind fields throughout the country.

Numerically simulated wind data is routinely used as input to statistical and machine learning models to obtain probabilistic forecasts for wind energy; see, e.g. Xie et al. (2022) for a comprehensive review. Relying entirely on numerical simulations would be computationally infeasible, as it would require an ensemble of simulations in real time for uncertainty quantification. As such, we aim at building a high-resolution spatio-temporal wind speed statistical model for prediction for the entire country which acts as a surrogate, as it is the practice in a vast body of work for other countries (Espeholt et al., 2022; Sobash & Ahijevych, 2024). Building such a statistical model is a very challenging task, given the considerable extension, as well as geographical diversity, of Saudi Arabia. Indeed, wind exhibits nonlinear dynamics that traditional statistical models, such as autoregressive integrated moving average (ARIMA), but also generalized additive models (Hastie, 2017), generalized multiplicative models (Atto et al., 2016), and generalized autoregressive conditional heteroskedasticity (GARCH, Bollerslev, 1986) cannot efficiently capture.

Dynamical models relying on neural networks (henceforth referred to as machine learning models) are increasingly applied in modelling time series as alternatives to conventional statistical models. Indeed, neural networks can be arranged in a recursive structure (Recurrent Neural Networks, RNNs) which naturally adapts to time series data and have been shown be able to capture nonlinear dynamics competitively or even better than traditional statistical models (Bonas, Datta, et al., 2024). RNNs are flexible, but their very large parameter space coupled with the recursive nature of the model imply an unstable evaluation of the gradient and hence a challenging implementation of gradient-based optimization (Doya, 1992). To address this issue, Jaeger (2001) proposed an Echo State Network (ESN), a RNN with a reduced parameter space implied by sparse and random matrices, a feature that greatly reduces the computational complexity for inference. A similar model where the matrix weights are also stochastic and sparse was proposed in the case where the input is (or can be converted to) spike trains (liquid-state machines, Bonas et al., 2025; Maass et al., 2002).

In the case of a relatively small number of locations, ESNs could be merged with covariance-based geostatistical models (Bonas & Castruccio, 2023), but larger data require some form of dimensional reduction in space. Recently, McDermott and Wikle (2017) and McDermott and Wikle (2019a, 2019b) applied ESNs (both shallow and deep) under both frequentist and Bayesian frameworks to model large geostatistical data with uncertainty quantification. In order to reduce the spatial dimensionality, these works adopt principal component analysis, a linear transformation aimed at finding the main modes of variability. As an alternative approach, Huang et al. (2022) proposed to select a collection of spatial knots, fit an ESN, forecast the values at the knots, and then interpolate the remaining field with some simple covariance-based geostatistical technique. Although representing an improvement, covariance-based interpolation (*kriging*) would still suffer from scalability issues with very large spatial data, such as the one in our application, which comprises wind speed across a country the size of one-fifth of the United States.

In this work, we propose a model that achieves spatial dimension reduction for ESN forecasting in a more accurate and efficient manner, and hence provides more timely and accurate wind speed forecasting at the lead times of interest for wind energy applications. Our proposed method finds a set of representative knots by means of an energy distance-based approach. Then, instead of

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performing covariance-based interpolation, the knots are modelled via a Stochastic Partial Differential Equation (SPDE). In this work, we rely on the equivalence of one of the most popular spatial models (the Matén model, Stein, 1999) to the solution from an advection-diffusion SPDE (Whittle, 1954), and generalize the differential operator to allow for spatially varying coefficients which can account for a nonstationary behaviour. The SPDE is discretized using a finite elements methods (Lindgren et al., 2011), which in this case is computationally convenient as the resulting model is a Gaussian Markov Random Field with sparse precision matrix. We also propose a new batch forecasting approach which is predicated on a sequential parameter re-estimation and allows for more accurate forecasting. Finally, inference is achieved using General Processing Unit (GPU) acceleration which will be shown to significantly improve the computational time.

The paper is organized as follows. Section 2 presents the wind speed simulations over Saudi Arabia. Section 3 introduces the trend, temporal and spatial structure of the model. Section 4 details how to perform inference. In Section 5 the model is applied to produce wind speed and energy forecasts. Section 6 concludes with a discussion.

2 Wind simulation data

In this work, we focus on hourly wind simulations from 2013 to 2016 in Saudi Arabia, using the Weather Research and Forecasting model (WRF; Skamarock et al., 2008), with boundary conditions from the high-resolution operational European Centre for Medium-Range Weather Forecast Model. These simulations were conducted on a regular planar grid, and were originally introduced in Giani et al. (2020) to assess the most suitable sites for wind farming.

In total, four simulations were performed, with resolutions at 9km and 6km and two different planetary boundary layer schemes. The simulated data were validated against observations collected from the ten sites of King Abdullah City for Atomic and Renewable Energy monitoring network. In this work, we focus on the simulation with the Mellor-Yamada-Janjić planetary boundary layer scheme (Gómez-Navarro et al., 2015; Janić, 2001; Mellor & Yamada, 1982) at 6km resolution, as this simulation was shown to be the closest to the actual observations (Giani et al., 2020). In addition, we only consider the wind simulations on the mainland, excluding the offshore islands, for a total of $n = 53{,}333$ locations. Every hour between 1 January 2013 and 31 December 2016 is considered, for a total of $T = 35{,}040$ time points. All together, this represents nearly 2 billions space-time locations.

The location-wise means and standard deviations of the wind speed data are shown in Figure 1. Figure 1a indicates that the wind speed tends to be higher around the mountain ranges in the west of Saudi Arabia. Figure 1b showcases how the wind speed is more variable in the mountain ranges close to the west coast. Wind speed in the rest of the country is lower and more stable in comparison due to a flat topography, especially in the empty quarter (Rub' al Khali) in the South East.

3 The spatio-temporal model

We define $Z_t(\mathbf{s}_i)$, where $i = 1, \ldots, n$ and $t = 1, \ldots, T$, as the wind speed observed at location $\mathbf{s}_i$ and time t. Section 3.1 provides a model for the trend. Section 3.2 explains how the data in space are dimensionally-reduced. Sections 3.3 and 3.4 provide the temporal and spatial model, respectively.

3.1 Trend model

We model hourly wind speed for multiple years with a spatially varying harmonic trend:

$$\sqrt{Z_t(\mathbf{s}_i)} = \beta_0(\mathbf{s}_i) + \sum_{p=1}^{P}\left\{\beta_{p,1}(\mathbf{s}_i)\cos\left(\frac{2\pi t}{T_p}\right) + \beta_{p,2}(\mathbf{s}_i)\sin\left(\frac{2\pi t}{T_p}\right)\right\} + \gamma(\mathbf{s}_i)Y_t(\mathbf{s}_i), \tag{1}$$

where P represents the number of harmonics. The parameter $\beta_0(\mathbf{s}_i)$, $i = 1, \ldots, n$, is the spatially varying intercept, while $\beta_{p,1}(\mathbf{s}_i)$ and $\beta_{p,2}(\mathbf{s}_i)$, $p = 1, \ldots, P$, denote the spatially varying coefficients for each harmonic pair at period T_p, and $\gamma(\mathbf{s}_i)$ is the location-wise normalizing parameter. The residual process $Y_t(\mathbf{s}_i)$ is assumed to be independent in time and with unit variance and zero mean. Following the preliminary analysis in Huang et al. (2022), we choose $P = 5$ with harmonics at

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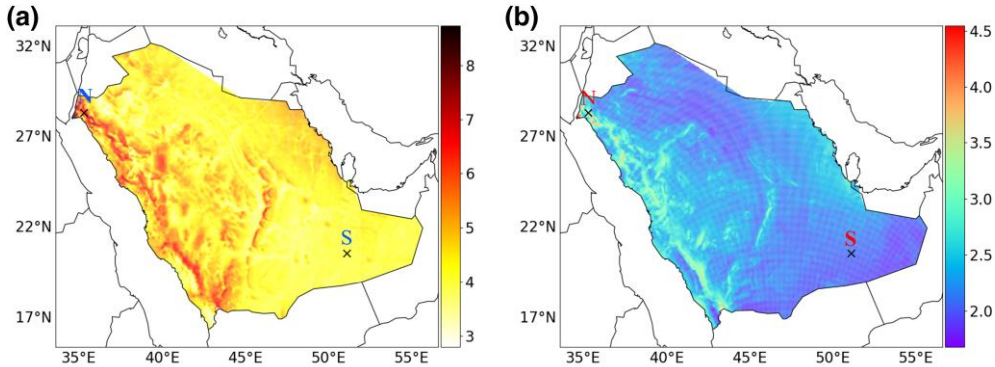

Figure 1. (a) Mean and (b) standard deviation of the WRF simulated hourly wind speed (m/s) over Saudi Arabia from 2013 to 2016. The N and S are the two sample locations (marked in black crosses) used to compute the diagnostics in online supplementary material Figure S1.

periods 1 year, 0.5 year, 1 day, 12 and 8 hr, so that $T_p = 4380$, 2920, 1460, 8, and 4, respectively. The square root of $Z_t(\mathbf{s}_i)$ allows us to obtain data that more closely resemble normality, as demonstrated in Figure S3 in the online supplementary material of Huang et al. (2022). The following sections focus on modelling the spatio-temporal structure of the wind speed residuals, $Y_t(\mathbf{s}_i)$.

3.2 Spatial dimension reduction

In general, it is impractical to model the entire vector of spatial wind speed residuals $Y_t(\mathbf{s}_i)$ for $n = 53{,}333$ locations. For the sake of modelling and scalability, it is convenient to reduce the spatial dimension n to $n_{\text{red}} \ll n$ ('red' stands for 'reduced') and in this work we rely on support points (Mak & Joseph, 2018) to select the most representative locations.

The support point method selects a set of locations that optimize the energy distance (Mak & Joseph, 2018; Székely & Rizzo, 2004) between an empirical and a prescribed distribution. Formally, we assume two distribution functions F and G with finite means and $\mathbf{S}, \mathbf{S}' \overset{\text{i.i.d.}}{\sim} G$ and $\mathbf{X}, \mathbf{X}' \overset{\text{i.i.d.}}{\sim} F$, and we consider the energy distance:

$$E(F, G) \equiv 2\mathbb{E}\|\mathbf{X} - \mathbf{S}\|_2 - \mathbb{E}\|\mathbf{S} - \mathbf{S}'\|_2 - \mathbb{E}\|\mathbf{X} - \mathbf{X}'\|_2.$$

Given that we have no prior information on the true distribution of the locations, we set $G = F_n$, where F_n is the empirical distribution function of F for $\{\mathbf{s}_i\}_{i=1}^n$. In other words, we assume that the original locations are a sample from F_n. The energy distance becomes

$$E(F, F_n) = \frac{2}{n}\sum_{i=1}^{n} \mathbb{E}\|\mathbf{X} - \mathbf{s}_i\|_2 - \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} \|\mathbf{s}_i - \mathbf{s}_j\|_2 - \mathbb{E}\|\mathbf{X} - \mathbf{X}'\|_2. \tag{2}$$

The support points are defined as the set of $n_{\text{red}} \ll n$ locations with resulting empirical distribution $F_{n_{\text{red}}}$, and are obtained by optimizing the energy distance:

$$\{\mathbf{s}_i\}_{i \in S_{\text{red}}} = \underset{\mathbf{x}}{\operatorname{argmin}}\, E(F_{n_{\text{red}}}, F_n), \tag{3}$$

where $\mathscr{S}_{\text{red}}$ is the set of indices for the selected locations ($|\mathscr{S}_{\text{red}}| = n_{\text{red}}$). The energy distance is non-negative for any two different distribution functions and is zero when the two distribution functions are equivalent. Therefore, by minimizing the energy distance we find the set of points for which the empirical distribution $F_{n_{\text{red}}}$ can best represent the true distribution F.

Song et al. (2025) theoretically proved that spatial interpolations based on support points converge to the true values at an optimal rate compared with those based on randomly or regularly spaced locations; the rationale behind the optimal convergence rate is that the locations selected

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with minimized energy distance to the empirical distribution of the locations can most efficiently represent the entire data, as will also be shown in the simulation studies in the online supplementary material. In this work, we apply support points to our $n = 53{,}333$ locations, and we aim at finding the set of locations $\{\mathbf{s}_i\}_{i \in \mathscr{S}_{\text{red}}}$, where $n_{\text{red}} = 3{,}173$ represents the number of selected knots (chosen to maintain consistency with the number of knots in Huang et al., 2022) to be optimized in equation (3), which can be achieved through sampling-based inference over the entire location set $\{\mathbf{s}_i\}_{i=1}^{n}$ (Mak & Joseph, 2018). To achieve spatial dimension reduction, we set $F = F_{n_{\text{red}}}$ in (3). The set of selected knots $(\mathbf{s}'_1, \ldots, \mathbf{s}'_{n_{\text{red}}})$ based on argmin is not necessarily a subset of the original locations. Therefore, we assign locations in the original set that are the closest to $(\mathbf{s}'_1, \ldots, \mathbf{s}'_{n_{\text{red}}})$ and denote this set as $\{\mathbf{s}_i\}_{i \in \mathscr{S}_{\text{red}}}$, the support points; see the online supplementary material for the optimality study and analysis of the support-point algorithm.

3.3 Temporal ESN model

In this section, we revisit the deep ESN model with $d = 1, \ldots, D$ layers introduced in Jaeger (2001) and then (McDermott & Wikle, 2017) to model the temporal dynamics of the wind speed residuals $\mathbf{Y}_t = (Y_t(\mathbf{s}_1), \ldots, Y_t(\mathbf{s}_{n_{\text{red}}}))^\top$. In the following sections, we propose a novel approach which uses ESN in conjunction with a spatial SPDE model, which allows for capturing more complex spatial dependence structure while still allowing for scalable and computationally affordable inference. The deep ESN is defined as:

$$\text{output:} \quad \mathbf{Y}_t = \sum_{d=1}^{D} \mathbf{B}_d k(\mathbf{h}_{t,d}) + \sum_{d=1}^{D} \mathbf{B}_d^2 k(\mathbf{h}_{t,d} \odot \mathbf{h}_{t,d}) + \boldsymbol{\epsilon}_t, \tag{4a}$$

$$\text{hidden state d:} \quad \mathbf{h}_{t,d} = (1-\alpha)\mathbf{h}_{t-1,d} + \alpha \omega_{t,d}, \tag{4b}$$

$$\omega_{t,d} = f\left(\frac{\nu_d}{|\lambda_d|}\mathbf{W}^d \mathbf{h}_{t-1,d} + \mathbf{W}^{d;\text{in}} \tilde{\mathbf{h}}_{t,d-1}\right), \quad \text{for } d > 1, \tag{4c}$$

$$\text{reduction } d-1: \quad \tilde{\mathbf{h}}_{t,d} = Q(\mathbf{h}_{t,d}), \quad \text{for } d > 1, \tag{4d}$$

$$\text{input:} \quad \omega_{t,1} = f\left(\frac{\nu_1}{|\lambda_{W_1}|}\mathbf{W}^1 \mathbf{h}_{t-1,1} + \mathbf{W}^{1;\text{in}} \mathbf{x}_t\right), \tag{4e}$$

$$\text{matrix distribution:} \quad W_{i,j}^d = \gamma_{i,j}^d G_{i,j}^d, \quad \gamma_{i,j}^d \sim \text{Be}(\pi_d),\ G_{i,j}^d \sim g(\eta_d), \tag{4 f}$$

$$W_{i,j}^{d;\text{in}} = \gamma_{i,j}^{d;\text{in}} G_{i,j}^{d;\text{in}}, \quad \gamma_{i,j}^{d;\text{in}} \sim \text{Be}(\pi_{d;\text{in}}),\ G_{i,j}^{d;\text{in}} \sim g(\eta_{d;\text{in}}), \quad g \text{ is a density symmetric about 0.} \tag{4g}$$

This nonlinear dynamic state-space model is such that in equation (4a) the n_{red}-dimensional vector $\mathbf{Y}_t$ is formulated as a linear combination of the state vectors $\mathbf{h}_{t,d}$ and of their quadratic states at each layer $d = 1, \ldots, D$. A scale location standardization $k(\cdot)$ is applied so that there is the same range of variability across layers. The error vector $\boldsymbol{\epsilon}_t$ is assumed to follow a Gaussian distribution with some covariance structure. The operator $\odot$ is the elementwise Hadamard product. The state vectors $\mathbf{h}_{t,d}$ are then expressed in equation (4b) as a convex combination with parameter α (the *leaking rate*) of the previous hidden state vector $\mathbf{h}_{t-1,d}$ and a vector $\boldsymbol{\omega}_{t,d}$ accounting for long-range dependence.

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In (4c), $\boldsymbol{\omega}_{t,d}$ is described as a combination of the state at the previous layer $\mathbf{h}_{t-1,d}$ weighted by a matrix $\mathbf{W}^d$ (the weight matrix at the dth layer) and a dimensionally reduced state vector $\tilde{\mathbf{h}}_{t,d-1}$ from the previous layer weighted by another matrix $\mathbf{W}^{d;\text{in}}$ (input weight matrix at the dth layer). The *activation function* $f(\cdot)$ in (4c) and (4e) is a hyperbolic tangent function, but other options such as the rectified linear or the sigmoid function could be used (Goodfellow et al., 2016). The matrices $\mathbf{W}^d$ are then divided by their largest eigenvalue λ_d to ensure the *echo-state property*, i.e. an asymptotic loss of dependence from the initial conditions (Jaeger, 2007; Lukoševičius, 2012). The parameter v_d allows us to scale the aforementioned matrix.

The dimension reduction for $\tilde{\mathbf{h}}_{t,d-1}$ is specified in (4d), through the function $Q(\cdot)$, extracting information from the state vector $\mathbf{h}_{t,d}$ as inputs for the next layer, and in this work we use principal component analysis. In the case of the first layer $d = 1$, equation (4e) explains how $\omega_{t,1}$ depends on a vector of previous observations $\mathbf{x}_t = (\mathbf{Y}_{t-\tau}, \ldots, \mathbf{Y}_{t-m\tau})^\top$, where τ denotes the forecast lead time, and m represents the number of past observations input (McDermott & Wikle, 2017).

The weight matrices $\mathbf{W}^d$ in (4c) and $\mathbf{W}^{d;\text{in}}$ in (4e) are both assumed to be random and sparse according to a spike-and-slab distribution as specified in Equations (4f) and (4g). Indeed, in both $\mathbf{W}^d$ and $\mathbf{W}^{d;\text{in}}$, each entry $W^d_{i,j}$ ($W^{d;\text{in}}_{i,j}$) is the outcome of a product between a Bernoulli random variable $\gamma^d_{i,j}$ ($\gamma^{d;\text{in}}_{i,j}$), respectively, times a random variable $G^d_{i,j}$ ($G^{d;\text{in}}_{i,j}$), which is drawn from a probability density function g symmetric about 0. The parameters η_d and $\eta_{d;\text{in}}$ control the shape of the distribution, in this case they represent the width of a zero-centred uniform distribution. The parameters π_d and $\pi_{d;\text{in}}$ control the matrix sparsity of $\mathbf{W}^d$ and $\mathbf{W}^{d;\text{in}}$ respectively.

In summary, except for the set of matrices $\mathbf{B} = \{\mathbf{B}_1, \ldots, \mathbf{B}_D\}$, the ESN has the following hyperparameters:

$$\boldsymbol{\theta} = (\{n_{h,d}, n_{\tilde{h},d}, v_d, \eta_d, \eta_{d;\text{in}}, \pi_d, \pi_{d;\text{in}}, d = 1, \ldots, D\}, m, \alpha, D).$$

3.4 Spatial SPDE model

In this section, we introduce the method used to model the spatial dependence structure of $Y_t(\mathbf{s}_i)$ at the n_{red} knots. As we assume that this process is independent and identically distributed in time, we drop the time index for simplicity and assume the existence of a continuously indexed random field $Y(\mathbf{s})$. We assume that this field is the solution of the following reaction-diffusion SPDE (Lindgren & Rue, 2015):

$$(\kappa^2(\mathbf{s}) - \Delta)^{\alpha/2}(\tau(\mathbf{s})Y(\mathbf{s})) = \mathrm{W}(\mathbf{s}), \quad \mathbf{s} \in \mathbb{R}^2, \quad \alpha = \nu + 1, \quad \kappa > 0, \quad \nu > 0, \tag{5}$$

where Δ denotes the Laplacian operator, and $\mathrm{W}(\mathbf{s})$ is a standard spatial Gaussian white noise process. If $\kappa^2(\mathbf{s}) = \kappa^2$ and $\tau(\mathbf{s}) = \tau$ are constant in space, the only stationary solution is a Gaussian random field with the Matérn correlation (Whittle, 1954), precision parameter τ, scale parameter κ, and smoothness ν (Stein, 1999). The marginal variance can be obtained as $\frac{\Gamma(\nu)}{\Gamma(\alpha)4\pi\kappa^{2\nu}\tau^2}$, and if ν is an integer, the discrete solution of the SPDE (5) is a Gaussian Markov random field (Rue & Held, 2005), a structure that implies sparse precision matrices and therefore fast likelihood evaluation and computationally convenient inference (Lindgren et al., 2011).

To allow for non-stationary spatial processes, Ingebrigtsen et al. (2014) proposed to formulate precision and scale parameters as linear combinations of basis functions:

$$\log(\tau(\mathbf{s})) = b^\tau_0(\mathbf{s}) + \sum_{o=1}^{p} b^\tau_o(\mathbf{s})\theta_o,$$

$$\log(\kappa(\mathbf{s})) = b^\kappa_0(\mathbf{s}) + \sum_{o=1}^{p} b^\kappa_o(\mathbf{s})\theta_o,$$

where $b^\tau_o(\cdot)$ and $b^\kappa_o(\cdot)$ are the Fourier basis functions with coefficient parameters θ_o.

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We assume prior independence among the coefficient parameters $b_o^\tau(\cdot)$ and $b_o^\kappa(\cdot)$ by assuming independent and identically distributed Gaussian priors with mean 0 and unit variance. We have also tried penalized complexity priors (Simpson et al., 2017) with 0.7 probability of the corresponding standard deviation (σ) being greater than 1 (μ) and we noticed minimal difference in the results, so we opted for the simpler choice.

4 Inference

4.1 Trend model

For the trend model (1), the coefficients $\beta_0(\mathbf{s}_i)$, $\beta_{o,1}(\mathbf{s}_i)$, $\beta_{o,2}(\mathbf{s}_i)$, and $\gamma(\mathbf{s}_i)$ are estimated by first assuming independence across space, and hence reducing the model for each location to a linear regression. As such, ordinary least squares can be performed on the entire wind data.

The residuals of the hourly model from 2013 to 2016 are then denoted as $\hat{Y}_t^{\text{trend}}(\mathbf{s}_i)$ and could be used to estimate the temporal and spatial dependence. Figure S1(a-b) online supplementary material shows the auto-correlation function for $\hat{Y}_t^{\text{trend}}(\mathbf{s}_i)$ for the two selected locations marked with black crosses in Figure 1. It is readily apparent how the temporal dependence is strong, and a model for the temporal structure is necessary.

4.2 Temporal ESN model

In order to perform inference on the ESN in equation (4), we first assume that the collection of matrices $\mathbf{B}$ is known, and we perform cross-validation by using wind speed data from 2013 to 2015 as a training set and 2016 as a testing set. The hyper-parameter vector $\boldsymbol{\theta}$ is then chosen as the minimizer of the mean squared error on the testing data. In order to estimate $\mathbf{B}$ we perform penalized least squares via ridge regression to mitigate overfitting. In symbols, we denote $\mathbf{Y} = \mathbf{HB} + \boldsymbol{\epsilon}$, with:

$$\mathbf{Y} = \begin{bmatrix} \mathbf{Y}_1^\top \\ \vdots \\ \mathbf{Y}_T^\top \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{h}_1 & , & \mathbf{h}_1^2 \\ \vdots & & \vdots \\ \mathbf{h}_T & , & \mathbf{h}_T^2 \end{bmatrix}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \boldsymbol{\epsilon}_1{}^\top \\ \vdots \\ \boldsymbol{\epsilon}_T^\top \end{bmatrix},$$

where $\mathbf{Y}_t = (\mathbf{Y}_t(\mathbf{s}_1), \ldots, \mathbf{Y}_t(\mathbf{s}_{n_{\text{red}}}))$. In addition, the state vector at time point t, $\mathbf{h}_t = \{k(\mathbf{h}_{t,D}), k(\tilde{\mathbf{h}}_{t,D-1}), \ldots, k(\tilde{\mathbf{h}}_{t,1})\}^\top$ and $\mathbf{h}_t^2 = \{k(\mathbf{h}_{t,D} \odot \mathbf{h}_{t,D}), k(\tilde{\mathbf{h}}_{t,D-1} \odot \tilde{\mathbf{h}}_{t,D-1}), \ldots, k(\tilde{\mathbf{h}}_{t,1} \odot \tilde{\mathbf{h}}_{t,1})\}^\top$, where $t = 1, \ldots, T$. Then, we estimate the collection of coefficient matrices $\hat{\mathbf{B}}$ via ridge regression is $(\mathbf{H}^\top\mathbf{H} + \lambda\mathbf{I})^{-1}\mathbf{H}^\top\mathbf{Y}$ for a penalty λ estimated via cross-validation.

To test the ability of our deep ESN model in capturing the temporal dynamics, we calculate the difference in L^2 norms of the auto-correlation function (up to 50 time lags) between $\hat{Y}_t^{\text{trend}}(\mathbf{s})$ and the residuals after the ESN predictions, which we denote as $\hat{Y}_t^{\text{trend+time}}(\mathbf{s})$. Figure S1(c-d) online supplementary material shows this improvement at the two sample locations (indicated by the crosses in Figure 1) and Figure S2 online supplementary material maps the difference in L^2 norms between the model with and without the temporal structure. It is readily apparent how there is a clear reduction with the medians (IQRs) of 1.110(0.307), 1.212(0.279), and 1.074(0.270) for the 3 lead hours, respectively.

4.3 Batch forecasting

Once the collection of matrices $\mathbf{B}$ has been estimated, the lead-one forecast at time point $T+1$ at the selected knots can be formulated as $\hat{\mathbf{Y}}_{T+1} = \hat{\mathbf{B}}^\top\mathbf{h}_{T+1}$. The long-lead forecasts at time $T+a$ can be obtained by iteratively using the predicted values $\hat{\mathbf{Y}}_{T+1}, \ldots, \hat{\mathbf{Y}}_{T+a-1}$ as part of the input vector $\mathbf{x}_t$. Instead of estimating $\mathbf{B}$ once, we propose a sequential update after a given batch window b. In practice, for all a of interest (in our application we have $a = 3$) we compute the forecast for time $T+a$ for all points $T+1, \ldots, T+b$, and then $\mathbf{B}$ is recomputed for the following batch. As will be shown in Sections 4.4 and 5, there is a trade-off in the choice of the window size b between computational efficiency and forecast accuracy, as a large b will result in fast but less accurate results. We henceforth denote this approach as *batch ESN (B-ESN)*.

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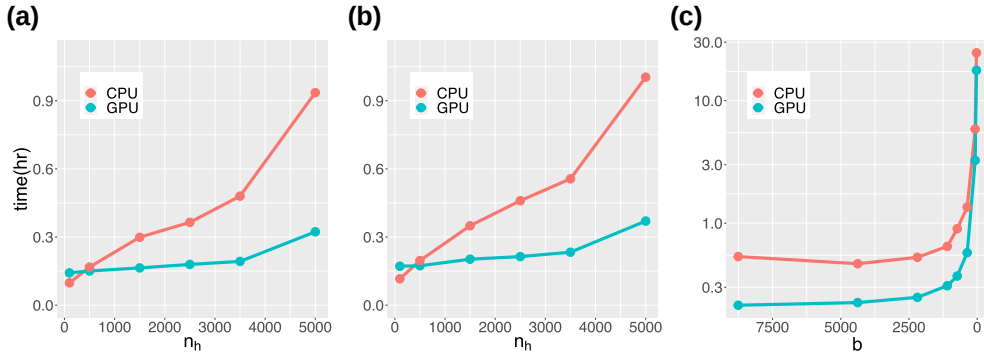

Figure 2. Computation time for inference and forecasting (3-hr ahead) of the ESN model (4) as a function of the number of hidden state dimension n_h for a small (panel (a)) and a large (b) number n_{red} of knots. Panel (c) show the trade-off between prediction efficiency and computation complexity as a function of the batch size for forecasting; the order of the *x*-axis is reversed to show the increasing trend of computation time against the frequency of updates. (a) $n_{\text{red}} = 100$, $b = 8{,}760$. (b) $n_{\text{red}} = 3,173$, $b = 8{,}760$. (c) $n_{\text{red}} = 3{,}173$.

4.4 Computational sensitivity

This section assessed the computational burden in the ESN model as a function of the number of hidden states n_h, number of locations n_{red}, the computing hardware (CPU, GPU) with the batch size b fixed at the temporal length of the test year 2016 to avoid updating the weight matrix $\hat{\mathbf{B}}$ during inference. As for the sensitivity analysis to the size of the forecast window, this section examines the computational complexity for varying values of b on the $n_{\text{red}} = 3{,}173$ knots selected in Section 3.2 in both the CPU and GPU computing environments. For both experiments, we used a 40-core Intel Cascade Lake machine with four V100 GPUs for this assessment. Figures 2a and 2b indicate that the hidden state dimension (n_h) has more impact on the computation complexity than the number of knots (n_{red}) due to the use of quadratic dynamics (5), for which the state vector $\mathbf{h}_t$ contains both linear and second-degree states. Therefore, the corresponding transition weight matrix $\mathbf{W}_d$ will scale as $O(4n_h^2)$. In contrast, as n_{red} increases from 100 to 3,173, the computation time does not change sensibly for the CPU and GPU computation time, which can be attributed to the $O(n_{\text{red}}^2)$ scaling rate of the associated input weight matrix $\mathbf{W}_d^{\text{in}}$. Also, the CPU computing time rapidly increases as n_{red} increases, while the GPU computing time is relatively constant. For large networks with $n_h = 5,000$, GPUs are considerably faster. If n_{red} is very small, the GPU will require more computational time due to input/output latency that outweighs any computational advantage.

Figure 2c shows the elapsed time for updating the weight matrix $\hat{\mathbf{B}}$ with differing batch sizes. It is immediately apparent that weight updates are very computationally expensive as updating frequency increases. Both CPU and GPU consume a significant amount of time as the updating frequency rises. The sharp increasing trend is due to the constant re-assembling of the hidden state matrix $\mathbf{H}$ and repeated re-estimating the coefficient matrix $\hat{\mathbf{B}}$. Consequently, although more frequent updates of the coefficient matrix are beneficial for the prediction accuracy of long-range forecasts, as indicated by various slopes of the MSPE curves for the 3 lead-hour forecasts shown in Figure 2c, time complexity must be considered in actual practice to balance the trade-off between prediction accuracy and efficiency. In an operational setting, the inference time for $\mathbf{B}$ cannot be comparable to the lead time for which the first forecast is sought. Figure 2c shows that when $b \approx 70$, the inference time for $\mathbf{B}$ is around 1 hr. Therefore, in order for the hourly forecasts to be meaningful, we maintain $b > 70$ throughout the application.

4.5 Spatial SPDE model

In order to perform inference on the spatial model, the SPDE (5) must be solved. The domain is therefore divided into a triangulated mesh containing non-intersecting triangles so that any two triangles on the mesh can have at most one overlapping corner or edge (Lindgren et al., 2011).

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We solve this SPDE by finite elements, assuming that the solution is of the form:

$$Y(\mathbf{s}) = \sum_{o=1}^{m} \psi_o(\mathbf{s}) Z_o, \tag{6}$$

where $\psi_o(\mathbf{s})$, $o = 1 \ldots, m$ are piecewise linear basis functions and m is the number of vertices formulated within the mesh. In addition, z_o are the associated weights following a Gaussian distribution. If v is an integer or half integer, $\mathbf{Z} = (Z_1, \ldots, Z_m)^\top$ is a Gaussian Markov random field whose precision matrix $\mathbf{\Sigma_Z}$ can then be approximated by the basis functions and their gradients. Moreover, $\mathbf{\Sigma_Z}$ involves a local precision parameter $\tau(\mathbf{s})$ and a scale parameter $\kappa(\mathbf{s})$, controlling uncertainty and scales of spatial correlations. The SPDE model can be summarized as:

$$\begin{aligned} \mathbf{Y} \mid \mathbf{z}, \sigma_e^2 &\sim \mathcal{N}_n(\mathbf{Az}, \sigma_e^2), \\ \mathbf{Z} &\sim \mathcal{N}_m(\mathbf{0}, \mathbf{\Sigma_Z}). \end{aligned} \tag{7}$$

In this model, $\mathbf{A}$ is the projection matrix from the latent spatial process to the locations of interest, allowing predictions on the entire domain. Here, $\mathbf{Z}$ represents the latent m-dimensional Gaussian Markov random field approximated by the piece-wise linear functions $\psi_o(\mathbf{s})$. The covariance $\mathbf{\Sigma_Z}$ describes the dependence of the process $\mathbf{Z}$, and since it is a Gaussian Markov random field, it allows for fast likelihood evaluation with a sparse precision matrix. The mesh grid is defined separately from the support points of the temporal model, whose observations are used for the estimation of the SPDE parameters.

4.6 Forecast calibration

Since the ESN assumes stochastic matrices, we can generate a forecast ensemble, with every member representing a different realization of the random matrices, whose uncertainty needs to be properly calibrated, so that a nominal 95% prediction interval would contain approximately 95% of the actual predictions. In this work, we adopt a calibration method as originally outlined in Bonas and Castruccio (2023), based on adjusting the variances under the assumption of Gaussianity. We first group the predictions according to the prediction horizon n_w and lead time k as follows:

$$\boldsymbol{f}_1^{(i)} = \begin{bmatrix} \hat{Y}_{T+1}^{T}(\mathbf{s}_i) \\ \hat{Y}_{T+2}^{T+1}(\mathbf{s}_i) \\ \vdots \\ \hat{Y}_{T+n_w+1}^{T+n_w}(\mathbf{s}_i) \end{bmatrix}, \quad \boldsymbol{f}_2^{(i)} = \begin{bmatrix} \hat{Y}_{T+2}^{T}(\mathbf{s}_i) \\ \hat{Y}_{T+3}^{T+1}(\mathbf{s}_i) \\ \vdots \\ \hat{Y}_{T+n_w+2}^{T+n_w}(\mathbf{s}_i) \end{bmatrix}, \ldots, \boldsymbol{f}_k^{(i)} = \begin{bmatrix} \hat{Y}_{T+k}^{T}(\mathbf{s}_i) \\ \hat{Y}_{T+k+1}^{T+1}(\mathbf{s}_i) \\ \vdots \\ \hat{Y}_{T+n_w+k}^{T+n_w}(\mathbf{s}_i) \end{bmatrix},$$

where $\hat{\mathbf{Y}}_{T+k}^{T} = \mathbb{E}(\mathbf{Y}_{T+k} \mid \mathbf{Y}_T, \ldots)$ and $i = 1, \ldots, n$ indexes the location. In our application, we predict values 3 hr ahead for the entire year 2016 by setting $k = 3$ and $n_w = 8,757$ (the number of hours in the testing year 2016). Then the lead forecasts can be grouped as $\boldsymbol{f}_1^{(i)} = \{\hat{Y}_{T+1}^{T}(\mathbf{s}_i), \hat{Y}_{T+3}^{T+2}(\mathbf{s}_i), \ldots, \hat{Y}_{T+n_w+1}^{T+n_w}(\mathbf{s}_i)\}$, $\boldsymbol{f}_2^{(i)} = \{\hat{Y}_{T+2}^{T}(\mathbf{s}_i), \hat{Y}_{T+5}^{T+3}(\mathbf{s}_i), \ldots, \hat{Y}_{T+n_w+2}^{T+n_w}(\mathbf{s}_i)\}$, and $\boldsymbol{f}_3^{(i)} = \{\hat{Y}_{T+3}^{T}(\mathbf{s}_i), \hat{Y}_{T+7}^{T+4}(\mathbf{s}_i), \ldots, \hat{Y}_{T+n_w+3}^{T+n_w}(\mathbf{s}_i)\}$. The residuals are then computed group-wise using the observed values as: $\mathbf{R}_j^{(i)} = \mathbf{Y}_j^{(i)} - \boldsymbol{f}_j^{(i)}$, $j = 1, 2, 3$, where $\mathbf{Y}_j^{(i)} = \{Y_{T+w}(\mathbf{s}_i); w = j, \ldots, n_w + j\}$ and $\mathbf{R}_j^{(i)} = \{R_j^{(i)}(w); w = j, \ldots, n_w + j\}$. The empirical standard deviation of the residuals $\mathbf{R}_j^{(i)}$ can be calculated as:

$$\hat{\sigma}_j^{(i)} = \sqrt{\frac{1}{n_w} \sum_{w=j}^{n_w+j} \left(R_j^{(i)}(w) - \bar{R}_j^{(i)} \right)^2}, \quad j = 1, 2, 3,$$

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where $\bar{R}_j^{(i)} = \sum_{w=j}^{n_w+j} R_j^{(i)}(w)/(n_w+1)$. If we formulate the calibrated residuals as $\tilde{\mathbf{R}}_{j,w} = (\tilde{R}_j^{(1)}(w), \ldots, \tilde{R}_j^{(n)}(w))$, then these vectors are assumed to follow a multivariate normal distribution as a consequence of the Gaussianity of the ESN in Section 4.2, i.e. $\tilde{\mathbf{R}}_{j,w} \overset{\text{i.i.d.}}{\sim} \mathcal{N}_n(\mathbf{0}, \mathbf{\Sigma})$ where independence is across $j = 1, 2, 3$ and $w = j, \ldots, n_w + j$. In order to estimate $\mathbf{\Sigma}$, we use a generalized shrinkage method (Castruccio et al., 2018; Friedman et al., 2008), which assumes that:

$$\hat{\mathbf{\Sigma}}^*(\delta) = \delta\hat{\mathbf{\Sigma}}_{\text{SPDE}} + (1-\delta)\hat{\mathbf{\Sigma}}_{\text{EMP}}, \tag{8}$$

where $\hat{\mathbf{\Sigma}}_{\text{SPDE}} = \mathbf{A}\hat{\mathbf{\Sigma}}_Z\mathbf{A}^\top$ as defined from the inference on the SPDE model (7), and $\hat{\mathbf{\Sigma}}_{\text{EMP}}$ is the empirical covariance matrix calculated from $\tilde{\mathbf{R}}_{j,w}$. The parameter $\delta \in (0, 1)$ is chosen to make prediction intervals for the spatial mean as close as possible to the expected coverage. This choice is justified by the fact that the variance of an average of dependent data is the most challenging quantity to estimate, as even a small misspecification of the covariance will propagate into this quantity. The covariance matrix $\hat{\mathbf{\Sigma}}^*(\delta)$ then characterizes the spatial correlation, which can be used to calculate the intervals for location-wise and aggregated forecasts.

More general penalized quantile regression approaches, as proposed in Bonas, Wikle, et al. (2024) and then in Huang et al. (2022) are possible, but only at the cost of an additional computational overhead which in this work was deemed excessive.

5 Application

5.1 Forecasting results

This section presents the forecasting results for our proposed model, considering the years 2013–2015 as the training set, and then 2016 as the testing set. The hourly wind predictions are performed on a rolling window basis, i.e. the forecast is continuously updated every hour.

Table 1 shows the MSPE across the knots (first part) and across all locations (second part) for lead time from 1 to 3 hr across a selection of models similar to that of the simulation study in the online supplementary material. In the first part of the table, only data at the knots are used, so no interpolation is necessary, while in the second part all locations are used (so the values are obtained through interpolation from the the knots). As in the simulation studies, our model is denoted as B-ESN (applying SPDE on B-ESN forecasts at the knots for global interpolations), while ESN denotes the one proposed in Huang et al. (2022), in both the cases of n_{red} knots and n locations. PER denotes the persistence approach, while B-ESN-L denotes our model but the spatial interpolation is performed with Lattice Kriging (Nychka et al., 2015). In the comparison study, we use two levels of multi-resolution basis to maintain the number of basis identical to the mesh vertices in the SPDE. Here GRU, LSTM, PER and VAR are the same models as in the simulation study in the online supplementary material.

Compared to ESN, our B-ESN model relying on support points, batch update ($b = 75$) and SPDE for interpolation results in improved forecasts for 2- and 3-hr lead prediction horizon in terms of both accuracy and stability. Indeed, our model provides the lowest medians and IQRs for these two forecasting horizons both at the knots and at all locations. In addition, the 1-hr lead forecasts obtained from B-ESN are very close to ESN for the knots and for all locations.

To better understand how B-ESN performs against ESN, Figure 3 maps the time-averaged relative forecasting error at all locations between these two models. Figure 3a demonstrates that the forecasts from the two models are very similar overall for 1-hr lead forecast, except in the Hejaz region (the western mountain ranges) where, however, wind farming is not practical due to logistics, construction costs and distance from the country's energy grid (Giani et al., 2020). Our B-ESN model outperforms the ESN for 2- and 3-hr ahead forecasts, as shown in Figures 3b and 3c, in which the relative error becomes almost uniformly smaller.

In both the case of knots and all locations, Table 1 shows that our B-ESN model significantly outperforms the other statistical and machine learning methods chosen: PER, VAR, LSTM, and GRU (the MPSE is higher across all locations because the evaluation is done over a larger number of points).

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Table 1. Predictive comparison in terms of median MSPE (computed across the temporal dimension) for different models (interquartile range or IQR in parenthesis)

Lead (hr)	One	Two	Three
Knots			
B-ESN	0.065 (0.214)	**0.115** (0.355)	**0.157** (0.470)
ESN	**0.064** (0.209)	0.125 (0.386)	0.173 (0.519)
GRU	0.308 (0.888)	0.380 (1.069)	0.428 (1.194)
LSTM	0.265 (0.743)	0.351 (0.951)	0.419 (1.127)
PER	0.077 (0.277)	0.198 (0.653)	0.311 (0.976)
VAR	0.087 (0.272)	0.162 (0.486)	0.216 (0.635)
All locations			
B-ESN	0.081 (0.267)	**0.133** (0.403)	**0.173** (0.510)
ESN	0.078 (0.255)	0.141 (0.426)	0.189 (0.557)
B-ESN-L	0.088 (0.290)	0.140 (0.425)	0.181 (0.530)
PER	**0.077** (0.276)	0.197 (0.669)	0.310 (0.974)

Note. Results for 1, 2, and 3 hr ahead forecasts at selected knots (first part of the table) and across all locations (second part) is shown. In bold is the name of our proposed approach and the minimum MSPE values across lead times. Median and IQR in time are used as the temporal distribution of GRU, LSTM, PER, and VAR are skewed.

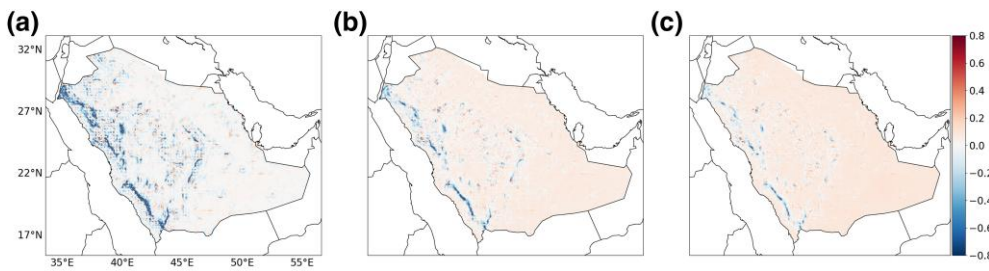

Figure 3. Location-wise relative forecasting errors between ESN and B-ESN for up to 3-hr lead ahead averaged across time. A positive number means that the B-ESN is better. (a) 1-hr lead. (b) 2-hr lead. (c) 3-hr lead.

5.2 Uncertainty quantification

This section details the results from the calibration approach proposed in Section 4.6. While marginally calibrating the forecast requires just a variance adjustment as indicated in Section 4.6, calibration of the joint distribution is a considerably more challenging task that in our model requires to choose the optimal convex combination between the covariance implied by the SPDE method and the empirical covariance, as shown in equation (8). Since uncertainty quantification is more challenging for averages across locations (even a small covariance misspecification would be magnified for the average), focussing on such a large domain with $n = 53,333$ locations is not feasible.

Therefore, instead of focussing on uncertainty quantification for the grand mean for the entire domain, we focus on calibration for an expanding square. We choose a square which expands following a small increment horizontally and vertically from a reference point near the centre of the domain, which in this case is 45° longitude and 20° latitude : $(45 - \Delta, 45 + \Delta) \times (20 - \Delta, 20 + \Delta)$, where $\Delta = 0.01, \ldots, 0.5$. We performed some preliminary tests of different initial locations and found comparable results. We then select the shrinkage parameter $\hat{\delta}(\Delta)$ that gives the best coverage for the grand mean of the forecasts within the expanding square, separately for each Δ. Finally, among all choices of $\hat{\delta}(\Delta)$, we choose the one which gives the most accurate marginal coverage: the median (across all locations in the domain) empirical coverage of a 95% needs to be as close as possible to the nominal 95% value. For the ease of notation, we denote this estimated parameter

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Table 2. Comparisons of 95%, 80%, and 60% location-wise marginal coverage computed from the calibrated covariance $\hat{\Sigma}^*(\hat{\delta})$, $\hat{\Sigma}_{\text{spatial}}$ the covariance estimated from the SPDE model, and the sample covariance $\hat{\Sigma}_{\text{EMP}}$

Lead One			
EXPECTED	$\hat{\Sigma}^*(\hat{\delta})$	$\hat{\Sigma}_{\text{spatial}}$	$\hat{\Sigma}_{\text{EMP}}$
0.95	0.949(0.022)	0.765(0.049)	0.977(0.013)
0.80	0.813(0.041)	0.569(0.042)	0.876(0.037)
0.60	0.628(0.047)	0.399(0.039)	0.698(0.043)
Lead Two			
EXPECTED	$\hat{\Sigma}^*(\hat{\delta})$	$\hat{\Sigma}_{\text{spatial}}$	$\hat{\Sigma}_{\text{EMP}}$
0.95	0.946(0.019)	0.800(0.045)	0.988(0.009)
0.80	0.809(0.039)	0.606(0.049)	0.906(0.030)
0.60	0.617(0.043)	0.431(0.040)	0.738(0.044)
Lead Three			
EXPECTED	$\hat{\Sigma}^*(\hat{\delta})$	$\hat{\Sigma}_{\text{spatial}}$	$\hat{\Sigma}_{\text{EMP}}$
0.95	0.940(0.022)	0.834(0.042)	0.992(0.005)
0.80	0.797(0.040)	0.646(0.048)	0.935(0.023)
0.60	0.604(0.047)	0.457(0.046)	0.780(0.045)

Note. The number here denotes the median of the empirical coverages across all locations and the number in parenthesis represents the IQR.

as $\hat{\delta}$, so with the same hat notation but without the dependence on Δ. Finally, this calibration is performed independently for each lead time, but again for the ease of notation we omit the dependence with respect to that.

Table 2 shows how the calibrated covariance $\hat{\Sigma}^*(\hat{\delta})$ can achieve the expected coverage with stability accurately. It is also readily apparent how $\hat{\Sigma}_{\text{spatial}}$ underestimates coverage, while $\hat{\Sigma}_{\text{EMP}}$ significantly overestimates it. This overestimation from $\hat{\Sigma}_{\text{EMP}}$ is due to the loss of local dependence structure assumed in the spatial model. Similarly, $\hat{\Sigma}_{\text{spatial}}$ underestimates the intervals because the assumption of the SPDE model on the entire study region, the entire country of Saudi Arabia in this case, is too strong. Hence, the dependence structure provided by $\hat{\Sigma}_{\text{spatial}}$ is not sufficient to account for variations on a global scale. Such observations are supported by the information provided in online supplementary material Table S3, demonstrating that $\hat{\delta}$ tends to decrease as the square expands (swinging from local to global dependence) and becomes very close to zero if the square is large enough. The convex combination of the spatial and sample covariance allows striking the balance between the regional spatial effects and the large-scale domain. It is also observable from online supplementary material Table S3 that as the forecast lead increases, the convex combination tends to favour the spatial covariance more. This result is expected because lead forecasts are generated based on predicted values at the previous time point from the ESN model; therefore, we depend increasingly on the spatial model to interpolate the field instead of the original data.

5.3 Wind power prediction

Electricity derived from wind power is challenging to store and must be consumed in a relatively short time. Therefore, ensuring accurate wind power forecasts is crucial to organizing the energy grid and estimating the additional electricity required from other energy sources in the near future. Wind power forecasts are typically focussed on short time horizons spanning from 1 to 3 hr, aiding in scheduling electricity transmission, allocating resources, and dispatching generated power (Gneiting et al., 2006; Hering & Genton, 2010; Zhu & Genton, 2012). In this section,

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Table 3. Annual sum of total absolute differences between the wind energy obtained using the forecasts from various models and true wind

Model	Absolute Energy Difference (KWh)
B-ESN	2.601×10^8
ESN	2.770×10^8
B-ESN-L	2.995×10^8
PER	3.120×10^8

we convert our wind speed forecasts into wind power and quantify the advantages from an operational point of view. Saudi Arabia does not yet have a developed energy market with pricing for each lead time, so as a reference, we focus on the wind power predictions based on the 2-hr lead forecasts.

To compute the wind power, an extrapolation from the wind at the surface (i.e. at a height of 10 m according to the WRF simulations) to the height of the turbine hub (80–120 m depending on the model) must be performed. Many approaches are available in the literature for vertical extrapolation of the wind at the hub height; see Gualtieri (2019) for a comprehensive review. The most popular approach assumes a power law relation at different heights as follows:

$$Z_t^{(h)}(\mathbf{s}) = Z_t(\mathbf{s})(h/10)^{\alpha(\mathbf{s})} \exp(\varepsilon_t), \tag{9}$$

where h denotes the hub height and $\alpha(\mathbf{s})$ is the *wind shear coefficient*, a parameter controlling the rate of extrapolation and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ is the error independent and identically distributed in time and space. The simplest approach is to assume that the wind shear coefficient is constant in space and fixed at 1/7, under the assumption of near-neutral atmospheric stability on a flat terrain (Gualtieri, 2019; Tagle et al., 2019). Recently, Crippa et al. (2021) proposed to estimate $\alpha(\mathbf{s})$, validated the proposed model against winds at the surface and at 40, 60, 80, and 100 m for the same data set that we are using here and demonstrated a significant improvement in the efficiency of wind power predictions. Therefore, here we use the same approach and fit the model in equation (9) by performing a linear regression for every pixel independently on the logarithmic scale. Once extrapolated at hub height, the wind speed is converted into energy via a power curve. Each turbine model has a different shape of the power curve, but shares the same structure: a cut-in speed (minimum value for which power is produced), a ramp-up, and a maximum rated output power until a cut-out speed for which the turbine is shut down to prevent damage; see online supplementary material Figure S9 for an example.

In this work, we focus on the 75 optimal locations for (inland) wind farms as indicated by Giani et al. (2020), which accounted for construction cost, maintenance, construction of power lines, and energy grid deployment; see online supplementary material Figure S10. For each of these locations, an optimal turbine model was identified, and here we will use the corresponding power curves for these models.

The annual sum of total absolute differences is computed and summarized in Table 3. It is readily apparent how the superior forecasting skills of B-ESN for wind speed against ESN, B-ESN-L and PER (the same models used in Table 1) translate into superior forecasting skills for wind power. This is also evident from Figure 4a, which shows the histogram of the absolute differences in wind energy of our B-ESN model against all aforementioned models.

Figure 4b demonstrates that the minimum annual sum of absolute differences is reached in the 50%–60% interval and the difference remains stable within the 50%–70% interval, implying that our combined model is resistant to overestimation of the wind speed. Moreover, as long as the forecast stays within the 20%–85% interval, increases in the annual sum of absolute differences are limited by 1×10^8 KWh, again testifying to the robustness of our model.

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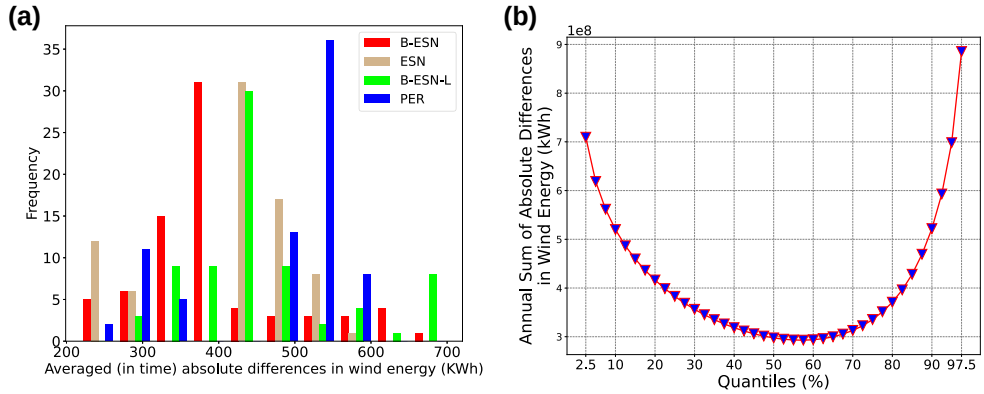

Figure 4. (a): Histograms of the absolute differences between the predicted wind power and the true wind power obtained from our B-ESN against ESN, B-ESN-L and PER. (b): Forecast quantiles against annual sum of the absolute differences between the B-ESN wind power predictions and the true wind power. (a) Model comparison. (b) Quantile comparison.

6 Conclusion

In this work, we propose a space-time model whose temporal part is controlled by an ESN, which is a sparse, stochastic RNN able to capture non-linear dynamics and long-range dependence for relatively short time series. Given the large size of the spatial data, dimension reduction is achieved via support points, which we showed in a simulation study to be able to better capture the spatial information than other sampling schemes. Finally, once the forecast is obtained, interpolation is performed using a non-stationary SPDE model, attaining overall higher prediction accuracy compared with previous approaches, as also shown in a simulation study. Overall, our proposed model allows for separate modelling of temporal and spatial dependence, and is more applicable to big datasets such as wind speed for a country as large as Saudi Arabia.

It should be pointed out that our model is designed as a tool for operational forecasting, as the boundary conditions for dynamical downscaling from the WRF simulation are provided by a reanalysis data product, not a forecasting data product. As such, for every time point we have no information about the future, and our statistical model is providing predictions using information only up to the present. Another potential application of our model would be as a stochastic weather generator trained on a dynamically downscaled ECMWF forecast to allow for uncertainty quantification. A direct application to a coarser regional model is also possible, although in that case the spatial component would probably not require dimension reduction via energy distance.

When fitting the spatial model, we fixed the smoothness parameter $\nu = 0.5$ to ensure the Gaussian Markov property for computational efficiency. Although Bolin and Kirchner (2020) have implemented the SPDE approach for any arbitrary values of ν through covariance-based rational approximations of fractional SPDEs (Bolin & Simas, 2023), applying this more general approach in this setting would likely lead to limited improvement in prediction accuracy because ν is estimated to be approximately 0.3 (close to 0.5) and the additional computational overhead of this approximation would hamper the efficiency of our operational forecasting approach, for which results must be provided in substantially less time than an hour.

Given the importance of accurate forecasts for wind power production, future work will investigate ensemble approaches to supplement the forecast proposed here with other competitive approaches. In particular, ongoing work is focussed on the use of (a) transformers and (b) convolutional auto-encoders as alternate means of spatial dimension reduction aimed at capturing more fine-scale details of wind across Saudi Arabia. Furthermore, better estimates of wind extrapolation to hub height (80–120 m) could be provided if the WRF simulations had sufficiently close vertical levels. For numerical stability, however, this would be possible only with a much finer spatial and temporal resolution, with a substantially higher computational burden. Finally,

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estimates of monetary savings would require additional context and an assessment of the uncertainty, and to this end ongoing work is focussed on characterizing and modelling the Saudi Arabia's energy market.

Finally, our promising results also motivate the development of a larger network to collect wind observational data in Saudi Arabia in the future.

Conflicts of interest: None declared.

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Data availability

The data are available through the KAUST library at the following link: https://repository.kaust.edu.sa/items/2187941f-4359-44b4-b593-1acd1e140d28. The code for this work can be found at the following GitHub repository: https://github.com/wangk0b/Deep-ESN-SPDE.

Supplementary material

Supplementary material is available online at *Journal of the Royal Statistical Society: Series C*.

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